Chapter-04

Measures of Central Tendency and Location

At the end of this chapter, students will be able to –

i. Define and explain the concept of central tendency and why summary statistics are essential for representing a dataset's typical value.

ii. Calculate the arithmetic mean, median, and mode accurately for both ungrouped and grouped (frequency distribution) data.

iii. Compare and contrast the strengths, weaknesses, and sensitivity of the mean, median, and mode in the presence of outliers and skewed distributions.

iv. Determine the most appropriate measure of central tendency based on the nature of the data and the level of measurement.

v. Compute weighted means and combined (pooled) means for multiple subgroups or combined datasets.

vi. Identify and apply the empirical relationships among the mean, median, and mode in moderately asymmetrical distributions.

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Measures of Central Tendency

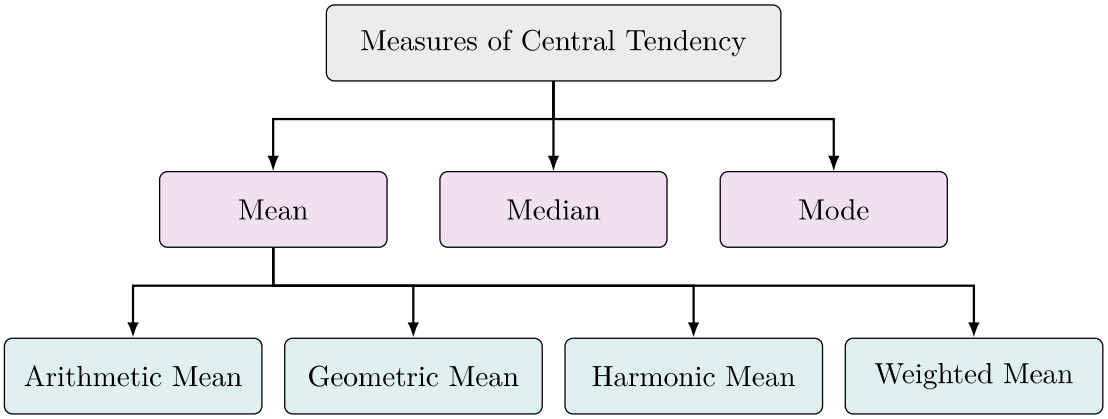
A measure of central tendency is a statistical value that describes the center, or typical value, of a data set. They summarize a whole distribution with a single number that represents where the data tends to cluster.

The most commonly used measures of tendency or averages are:

1. Mean
2. Median
3. Mode

Less common measures include:

- Arithmetic Mean
- Geometric Mean
- Harmonic Mean
- Weighted Mean



Characteristics of a Good Measure of Central Tendency

A good measure of location or central tendency, often described by Yule's criteria, should have the following characteristics:

i. It should be rigidly defined.

ii. It should be easy to understand and calculate.

iii. It should be based on all observations.

iv. It should not be affected by extreme values.

v. It should be capable of further algebraic treatment.

vi. It should have sampling stability.

vii. It should be suitable for open-ended classes.

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4.1. Arithmetic mean

4.1.1 Arithmetic Mean for Ungrouped Data

The arithmetic mean is the sum of all observations divided by the total number of observations.

For observations $x_1, x_2, \ldots, x_n$, the arithmetic mean is defined as

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

where

- $\bar{x}$ = arithmetic mean
- $\sum x_i$ = sum of all observations
- n = number of observations

Example: Find the mean of $5, 8, 12, 15, 20$.

$$\bar{x} = \frac{5 + 8 + 12 + 15 + 20}{5} = \frac{60}{5} = 12$$

4.1.2 Population mean

The population mean is the arithmetic mean of all the values in a population. It is a fixed value (a parameter) that describes the center of the whole population, and it is denoted by the Greek letter μ (mu).

Formula: For a population of N values $x_1, x_2, \ldots, x_N$,

$$\mu = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{\sum_{i=1}^{N} x_i}{N}$$

where

- μ = population mean,
- x_i = the i-th value in the population,
- N = total number of values in the population.

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Example: A class has 5 students with marks $60, 70, 75, 80, 90$. Treating these 5 students as the whole population,

$$\mu = \frac{60 + 70 + 75 + 80 + 90}{5} = \frac{375}{5} = 75$$

Note: The population mean μ is a parameter. The mean of a sample, $\bar{x}$, is a statistic used to estimate μ. The population mean uses N for the number of values, while the sample mean uses n.

4.1.3 Arithmetic mean for grouped data or frequency distribution

When the same value occurs more than once, the data are arranged as a frequency distribution. Each value x_i is multiplied by its frequency f_i, and the products are added and divided by the total frequency.

Formula: For k classes with midpoints $x_1, x_2, \ldots, x_k$ and frequencies $f_1, f_2, \ldots, f_k$,

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \cdots + f_k x_k}{f_1 + f_2 + \cdots + f_k} = \frac{\sum_{i=1}^{k} f_i x_i}{\sum_{i=1}^{k} f_i}$$

where

- $\bar{x}$ = arithmetic mean,

- x_i = midpoint of the i-th class, $x_i = \dfrac{\text{lower limit} + \text{upper limit}}{2}$,
- f_i = frequency of the i-th class,
- $\sum f_i = n$ = total frequency.

One can compute $\bar{x}$ by using the following table:

Class	Frequency (f_i)	Midpoint (x_i)	$f_i x_i$
L_1–U_1	f_1	x_1	$f_1 x_1$
L_2–U_2	f_2	x_2	$f_2 x_2$
L_3–U_3	f_3	x_3	$f_3 x_3$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
L_k–U_k	f_k	x_k	$f_k x_k$
Total	$\sum f_i = n$		$\sum_{i=1}^{k} f_i x_i$

Here L_i and U_i are the lower and upper limits of the i-th class, and $x_i = \dfrac{L_i + U_i}{2}$. Then

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}.$$

Example: Find the mean of the following distribution.

Class	Frequency (f_i)	Midpoint (x_i)	$f_i x_i$
0–10	5	5	25
10–20	8	15	120
20–30	15	25	375
30–40	16	35	560
40–50	6	45	270
Total	$n = 50$		$\sum f_i x_i = 1350$

$$\bar{x} = \frac{\sum f_i x_i}{n} = \frac{1350}{50} = 27$$

Note: For grouped data, every value in a class is assumed to be equal to the class midpoint, so the result is an approximation of the true mean. For a discrete frequency distribution (single values instead of classes), the same formula is used with x_i as the actual value.

Example: Find the arithmetic mean of the following distribution by (i) the direct method and (ii) the short-cut method.

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Class	Frequency (f)
0–10	5
10–20	8
20–30	15
30–40	16
40–50	6
Total	$n = 50$

Solution:

(i) Direct Method

Class	Frequency (f)	Midpoint (x)	fx
0–10	5	5	25
10–20	8	15	120
20–30	15	25	375
30–40	16	35	560
40–50	6	45	270
Total	$\sum f = 50$		$\sum fx = 1350$

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$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1350}{50} = 27$$

(ii) Short-Cut Method

In this method a convenient value A is taken as the assumed mean, and the deviation of each midpoint from it is used. This keeps the products small, so the calculation is easier.

Formula:

$$\bar{x} = A + \frac{\sum_{i=1}^{k} f_i d_i}{\sum_{i=1}^{k} f_i} \times h \qquad \text{where } d_i = \frac{x_i - A}{h}.$$

Solution: Take the assumed mean $A = 25$, which is the midpoint of a class near the middle of the data.

Class	Frequency (f)	Midpoint (x)	$d_i = \dfrac{x-25}{h}$	$f_i d_i$
0–10	5	5	-2	-10
10–20	8	15	-1	-8
20–30	15	25	0	0
30–40	16	35	1	16
40–50	6	45	2	12
Total	$n = 50$			$\sum f_i d_i = 10$

$$\bar{x} = A + \frac{\sum f_i d_i}{n} \times h = 25 + \frac{10}{50} \times 10 = 25 + 2 = 27$$

All three methods give the same answer, $\bar{x} = 27$.

Why the short-cut method works?

Since $d_i = \dfrac{x_i - A}{h}$, we have $x_i = A + h\, d_i$. Hence

$$\sum f_i x_i = \sum f_i (A + h\, d_i) = A \sum f_i + h \sum f_i d_i$$

Dividing by $\sum f_i$,

$$\bar{x} = A + h\, \frac{\sum f_i d_i}{\sum f_i}$$

which is the step-deviation formula.

Note:

- Any value can be chosen as the assumed mean A, and the final answer does not change. Choosing a midpoint near the center of the data keeps the deviations small.
- The answer is an approximation of the true mean, because every value in a class is taken to equal the class midpoint.

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4.1.4 Weighted mean

The weighted mean is an average in which each observation is given a weight according to its importance. It is used when the values do not all contribute equally to the result, for example marks in an exam where the final test counts more than a quiz.

Formula: For values $x_1, x_2, \ldots, x_n$ with corresponding weights $w_1, w_2, \ldots, w_n$,

$$\bar{x}_w = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_n x_n}{w_1 + w_2 + \cdots + w_n} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

where

- $\bar{x}_w$ = weighted mean,
- x_i = the i-th value,
- w_i = the weight of the i-th value.

One can compute $\bar{x}_w$ by using the following table:

Value (x_i)	Weight (w_i)	$w_i x_i$
x_1	w_1	$w_1 x_1$
x_2	w_2	$w_2 x_2$
$\vdots$	$\vdots$	$\vdots$
x_n	w_n	$w_n x_n$
Total	$\sum_{i=1}^{n} w_i$	$\sum_{i=1}^{n} w_i x_i$

Example: A student scores 80 in an assignment (weight 2), 70 in a midterm (weight 3) and 90 in the final exam (weight 5). Find the weighted mean.

Marks (x)	Weight (w)	wx
80	2	160
70	3	210
90	5	450
Total	$\sum w_i = 10$	$\sum w_i x_i = 820$

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i} = \frac{820}{10} = 82$$

Note:

- If all weights are equal, the weighted mean is the same as the simple arithmetic mean. In the example above, the simple mean is $\frac{80+70+90}{3} = 80$, which differs from 82 because the final exam has the largest weight.
- The arithmetic mean for a frequency distribution is a weighted mean in which the frequencies f_i are the weights.

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4.1.5 Merits of arithmetic mean

It is the most popular and widely used average in practice. It has the following merits

1. It is rigidly defined by a mathematical formula.
2. It is easy to understand and calculate.
3. It is based on all observations.
4. It is capable of further algebraic treatment.
5. It is least affected by sampling fluctuations.
6. It is the most widely used measure of central tendency.

4.1.6 Demerits of arithmetic mean

It is the best measures of average among all the averages. However, it has some limitations too.

1. It is highly affected by extreme values (outliers).
2. It cannot be calculated for open-ended classes without an assumption.
3. It cannot be found graphically.
4. It is not suitable for qualitative data.
5. It may give a misleading result for skewed data.
6. It may not be an actual value in the data set.

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4.2. Median

The median is the middle value of a data set when the observations are arranged in ascending (or descending) order. It divides the data into two equal parts, so that 50% of the observations are below it and 50% are above it.

4.2.1 Median for ungrouped data

Procedure:

1. Arrange the n observations in ascending order.
2. If n is odd, the median is the middle value.
3. If n is even, the median is the average of the two middle values.

Formula:

- If n is odd, the median is defined as

$$\text{Median} = \left(\frac{n+1}{2}\right)\text{-th observation.}$$

- If n is even, the median is defined as

$$\text{Median} = \frac{\left(\frac{n}{2}\right)\text{-th observation} + \left(\frac{n}{2}+1\right)\text{-th observation}}{2}.$$

Example 1 (n odd): Find the median of $12, 5, 9, 20, 7$.
Ascending order: $5, 7, 9, 12, 20$. Here $n = 5$, so

$$\text{Median} = \left(\frac{5+1}{2}\right)\text{-th observation} = 3\text{rd observation} = 9$$

Example 2 (n even): Find the median of $4, 10, 6, 8, 12, 2$.
Ascending order: $2, 4, 6, 8, 10, 12$. Here $n = 6$, so the middle values are the 3rd and 4th values.

$$\text{Median} = \frac{6+8}{2} = 7$$

4.2.2 Median from grouped data of a discrete variable

When the data are given as a frequency distribution of a discrete variable, the median is found from the cumulative frequencies.

Procedure:

1. Arrange the values x_i in ascending order and find the total frequency $n = \sum f_i$.
2. Compute the cumulative frequency (cf) of each value.
3. If n is odd, locate the $\left(\frac{n+1}{2}\right)$-th observation. The median is the value x_i whose cumulative frequency first becomes greater than or equal to this position.
4. If n is even, locate the $\left(\frac{n}{2}\right)$-th and $\left(\frac{n}{2}+1\right)$-th observations, find the values x_i they belong to, and take their average.

One can compute the median by using the following table:

Value (x_i)	Frequency (f_i)	Cumulative frequency (cf)
x_1	f_1	f_1
x_2	f_2	$f_1 + f_2$
x_3	f_3	$f_1 + f_2 + f_3$
$\vdots$	$\vdots$	$\vdots$
x_k	f_k	n
Total	$\sum_{i=1}^{k} f_i = n$	

Example 3 (n odd): Find the median of the following distribution.

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Value (x_i)	Frequency (f_i)	Cumulative frequency (cf)
1	3	3
2	7	10
3	12	22
4	8	30
5	5	35
Total	$n = 35$	

Here $n = 35$, so we need the $\dfrac{35+1}{2} = 18$th observation. The first cumulative frequency that is at least 18 is 22, which belongs to $x = 3$. Hence

$$\text{Median} = 3$$

Example 4 (n even): Find the median of the following distribution.

Value (x_i)	Frequency (f_i)	Cumulative frequency (cf)
10	4	4
20	6	10
30	6	16
40	4	20
Total	$n = 20$	

Here $n = 20$, so we need the $\dfrac{20}{2} = 10$th and $\left(\dfrac{20}{2}+1\right) = 11$th observations. The 10th observation lies in the class with $cf = 10$, so it is 20. The 11th observation lies in the class with $cf = 16$, so it is 30. Hence

$$\text{Median} = \frac{20+30}{2} = 25$$

Note:

- The data must always be arranged in order before the median is found.
- The median is not affected by extreme values, so it is a good measure for skewed data.
- In Example 4 the median (25) is not one of the observed values. When the two middle observations differ, the median can fall between values.

4.2.3 Median from grouped data of continuous variable

When the data of a continuous variable are grouped into classes, the exact median cannot be read from the table. It is estimated by finding the class that contains the median (the *median class*)

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and then interpolating within that class. This assumes that the observations in the median class are spread evenly across it.

Formula:

$$\text{Median} = L + \frac{\frac{n}{2} - f_c}{f_m} \times h$$

where

- L = lower boundary of the median class,
- $n = \sum f_i$ = total frequency,
- f_c = cumulative frequency of the class just before the median class,
- f_m = frequency of the median class,
- h = width (size) of the median class.

Procedure:

1. Write the classes in ascending order and find $n = \sum f_i$.
2. Compute the cumulative frequency (cf) of each class.
3. Find $\frac{n}{2}$.
4. The median class is the first class whose cumulative frequency is greater than or equal to $\frac{n}{2}$.
5. Read off L, f_c, f_m and h for that class and substitute them in the formula.

One can compute the median by using the following table:

Class	Frequency (f_i)	Cumulative frequency (cf)
L_1–U_1	f_1	f_1
L_2–U_2	f_2	$f_1 + f_2$
L_3–U_3	f_3	$f_1 + f_2 + f_3$
$\vdots$	$\vdots$	$\vdots$
L_k–U_k	f_k	n
Total	$\sum f_i = n$	

Example: Find the median of the following distribution.

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Class	Frequency (f)	Cumulative frequency (cf)
0–10	5	5
10–20	8	13
20–30	15	28
30–40	16	44
40–50	6	50
Total	$n = 50$	

Here $n = 50$, so $\frac{n}{2} = 25$. The first cumulative frequency that is at least 25 is 28, so the median class is 20–30. Therefore

$$L = 20, \qquad f_c = 13, \qquad f_m = 15, \qquad h = 10$$

and

$$\begin{aligned} \text{Median} &= L + \frac{\frac{n}{2} - f_c}{f_m} \times h \\ &= 20 + \frac{25 - 13}{15} \times 10 \\ &= 20 + \frac{12}{15} \times 10 \\ &= 20 + 8 \\ &= 28. \end{aligned}$$

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Note:

- The classes must be continuous (exclusive), such as 0–10, 10–20, with no gaps. If the classes are inclusive, such as 0–9, 10–19, first convert them to class boundaries by subtracting 0.5 from each lower limit and adding 0.5 to each upper limit.
- The result is an estimate, because the exact positions of the values inside the median class are not known.
- For open-ended classes the median can still be found, provided the median class itself is not open-ended.

Locating the median graphically

The median of grouped data of a continuous variable can also be found from a graph. Two methods are used: the ogive (cumulative frequency curve) and the histogram. Both give the same estimate as the interpolation formula. The same distribution is used for all the examples below.

Class	Frequency (f)
0–10	5
10–20	8
20–30	15
30–40	16
40–50	6
Total	$n = 50$

4.2.4 Median from the ogive curve

An ogive is a graph of cumulative frequencies plotted against class boundaries. There are two types: the "less than" ogive and the "more than" ogive.

Method 1: Using the "less than" ogive

Procedure:

1. Prepare the "less than" cumulative frequency table.
2. Plot the upper class boundaries on the x-axis and the cumulative frequencies on the y-axis. Start the curve at the lower boundary of the first class with cumulative frequency 0.
3. Join the points by a smooth curve (or straight lines). This is the ogive.
4. Mark $\frac{n}{2}$ on the y-axis.
5. Draw a horizontal line from $\frac{n}{2}$ until it meets the ogive.
6. From that point draw a vertical line down to the x-axis. The value where it meets the x-axis is the median.

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Class Boundary	Less Than Cumulative frequency
Less 0	0
Less 10	5
Less 20	13
Less 30	28
Less 40	44
Less 50	50

Here $\frac{n}{2} = \frac{50}{2} = 25$. The horizontal line at 25 meets the ogive above $x = 28$, so

$$\text{Median} = 28.$$

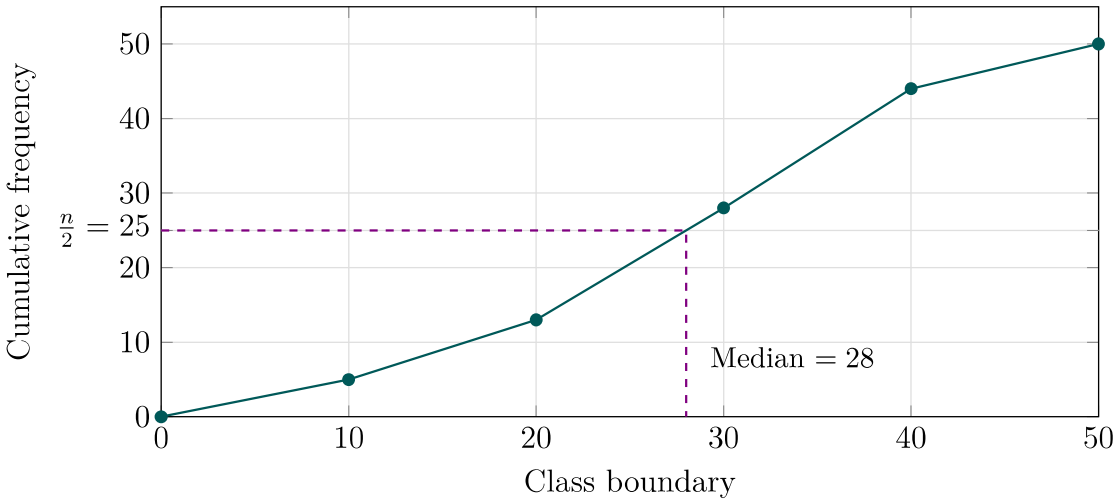


Figure 9: Median from the "less than" ogive curve.

Method 2: Using both ogives

The median is also the x-coordinate of the point where the "less than" and "more than" ogives intersect.

Procedure:

1. Prepare both the "less than" and the "more than" cumulative frequency tables.
2. Plot both ogives on the same graph, using the same axes.
3. Mark the point where the two curves cross.
4. Draw a vertical line from the intersection point to the x-axis. This value is the median.

Class Boundary	More than Cumulative frequency
More 0	50
More 10	45
More 20	37
More 30	22
More 40	6
More 50	0

The curves intersect at $(28, 25)$, so the median is 28. The y-coordinate of the intersection is always $\frac{n}{2}$.

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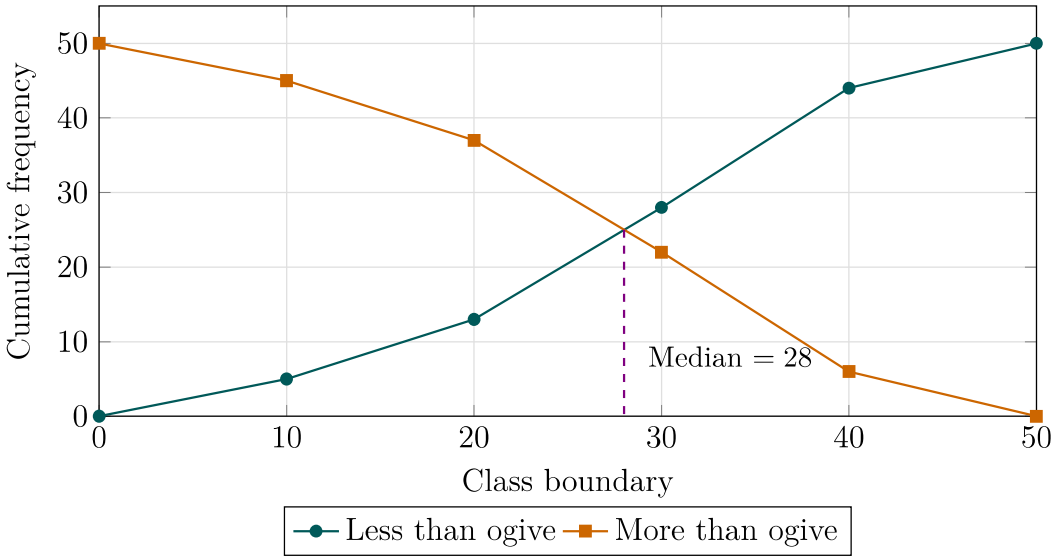


Figure 10: Median from the intersection of the two ogive curves.

4.2.5 Median from the histogram

The median divides the total area of the histogram into two equal parts. So the median is the value on the x-axis where a vertical line splits the histogram into two parts of equal area.

Procedure:

1. Draw the histogram, with class boundaries on the x-axis and frequencies on the y-axis.
2. Find $\frac{n}{2}$ and identify the median class, as in the formula method.
3. The vertical line lies inside the median class rectangle. The area to its left must equal $\frac{n}{2}$, so within the median class it must cover a frequency of $\frac{n}{2} - f_c$, where f_c is the cumulative frequency before the median class.
4. Mark this width inside the median class rectangle and draw the vertical line there. The point where it meets the x-axis is the median.

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Here the median class is 20–30. The classes before it hold $f_c = 13$, so the line must take up $25 - 13 = 12$ more units of frequency out of the 15 in this bar. This is $\frac{12}{15}$ of the bar width, which is $\frac{12}{15} \times 10 = 8$. Hence the line is drawn at $20 + 8 = 28$, and

$$\text{Median} = 28.$$

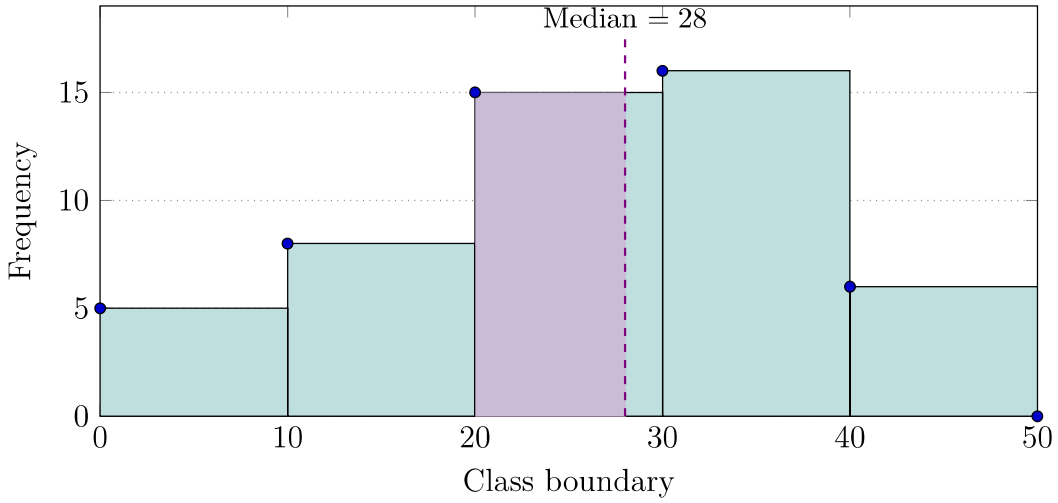


Figure 11: Median from the histogram (shaded part is the median class area to the left of the median)

Note:

- The graphical result is an estimate, and its accuracy depends on how carefully the graph is drawn. The formula gives the exact interpolated value.
- For a histogram with unequal class widths, plot the frequency density $\left(\frac{f}{h}\right)$ on the y-axis, so that the area of each bar is proportional to its frequency.
- In the ogive, the classes must be continuous (exclusive). If they are inclusive, first convert them to class boundaries.

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4.2.6 Merits of median

1. It is rigidly defined.
2. It is easy to understand and calculate.
3. It is not affected by extreme values (outliers).
4. It can be calculated for open-ended classes.
5. It can be located graphically, from the ogive or the histogram.
6. It is suitable for skewed data.
7. It can be used for qualitative data that can be ranked, such as grades.

4.2.7 Demerits of median

1. It is not based on all observations.
2. It is not capable of further algebraic treatment.
3. The data must be arranged in order before it can be found.
4. It is more affected by sampling fluctuations than the mean.

5. For grouped data, it is only an estimate based on the assumption of even spread within the median class.
6. It cannot be found exactly for a very large data set without sorting.

4.3. Mode

The mode is the value that occurs most frequently in a data set. It is the most common or most typical value. A data set may have one mode (unimodal), two modes (bimodal), more than two modes (multimodal), or no mode at all if no value is repeated.

4.3.1 Mode for ungrouped data

Procedure: Count how many times each value occurs. The value with the highest count is the mode.

Example 1 (one mode): Find the mode of $3, 5, 7, 5, 8, 5, 9$.
The value 5 occurs three times and every other value occurs once. Hence

$$\text{Mode} = 5$$

Example 2 (two modes): Find the mode of $2, 3, 3, 4, 5, 5, 6$.
The values 3 and 5 each occur twice, which is more than any other value. The data are bimodal, with

$$\text{Modes} = 3 \text{ and } 5$$

Example 3 (no mode): In $1, 2, 3, 4, 5$ every value occurs once, so there is no mode.

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4.3.2 Mode for qualitative data

The mode is the only measure of central tendency that can be used for qualitative (categorical) data, because the mean and median need numerical values or a ranking. The mode is the category with the highest frequency.

Example: The shirt sizes sold in a shop to 60 customers were recorded as follows.

Shirt Size	**Tally Marks**	**Frequency**
Small (S)	𝍸 𝍸 𝍸	15
Medium (M)	𝍸 𝍸 𝍸 𝍸 𝍸 |	26
Large (L)	𝍸 𝍸 ||	12
Extra Large (XL)	𝍸 ||	7
Total		**60**

Here $n = 60$. The frequencies are 15, 26, 12 and 7. The highest frequency is 26, which belongs to Medium (M). Hence

$$\text{Mode} = \text{Medium (M)}$$

That is, Medium is the most commonly sold shirt size, bought by $\frac{26}{60} \approx 43.3\%$ of the customers.



Figure 12: Bar chart of shirt sizes. The tallest bar (M) is the mode.

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Note:

- The mode of qualitative data is a category (here, Medium), not a number.
- The mode is the category with the highest frequency, not the highest frequency itself. The frequency 26 is not the mode.
- If two categories tie for the highest frequency, the data are bimodal and both categories are modes.

4.3.3 Mode for a discrete frequency distribution

For a frequency distribution of a discrete variable, the mode is the value x_i that has the highest frequency f_i.

One can find the mode by using the following table:

Value (x_i)	Frequency (f_i)
x_1	f_1
x_2	f_2
$\vdots$	$\vdots$
x_k	f_k
Total	$\sum_{i=1}^{k} f_i = n$

Example: Find the mode of the following distribution.

x	f
1	3
2	7
3	12
4	8
5	5
Total	$n = 35$

The highest frequency is 12, which belongs to $x = 3$. Hence

$$\text{Mode} = 3$$

4.3.4 Mode from grouped data of continuous variable

When the data of a continuous variable are grouped into classes, the mode is estimated by finding the class with the highest frequency (the *modal class*) and then interpolating within it.

Formula:

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

where

- $\Delta_1 = f_1 - f_0$ and $\Delta_2 = f_1 - f_2$
- L = lower boundary of the modal class,
- f_1 = frequency of the modal class,
- f_0 = frequency of the class just before the modal class,
- f_2 = frequency of the class just after the modal class,
- h = width (size) of the modal class.

Procedure:

1. Write the classes in ascending order with their frequencies.
2. Find the modal class, which is the class with the highest frequency.
3. Read off L, f_0, f_1, f_2 and h and substitute them in the formula.

One can compute the mode by using the following table:

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Class	Frequency (f_i)
L_1–U_1	f_1
L_2–U_2	f_2
$\vdots$	$\vdots$
L_k–U_k	f_k
Total	$\sum_{i=1}^{k} f_i = n$

Example: Find the mode of the following distribution.

Class	Frequency (f)
0–10	5
10–20	8
20–30	15
30–40	16
40–50	6
Total	$n = 50$

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The highest frequency is 16, so the modal class is 30–40. Therefore

$$L = 30, \qquad f_1 = 16, \qquad f_0 = 15, \qquad f_2 = 6, \qquad h = 10$$

and

$$\begin{aligned}
\text{Mode} &= L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h \\
&= 30 + \frac{16 - 15}{2(16) - 15 - 6} \times 10 \\
&= 30 + \frac{1}{11} \times 10 \\
&\approx 30.91
\end{aligned}$$

Note:

- The formula is reliable when the classes are continuous (exclusive) and have equal widths. For inclusive classes, such as 0–9, 10–19, first convert them to class boundaries.
- The result is an estimate, because the exact values inside the modal class are not known.
- For a moderately skewed distribution, the empirical relation

$$\text{Mode} \approx 3\,\text{Median} - 2\,\text{Mean}$$

gives a quick check. For this data the mean is 27 and the median is 28, so the relation gives $3(28) - 2(27) = 30$, which is close to 30.91.

Locating the mode graphically

The mode of grouped data of a continuous variable can be found from the histogram. The method uses the modal class bar and the two bars on either side of it. The same distribution as before is used here.

Class	Frequency (f)
0–10	5
10–20	8
20–30	15
30–40	16
40–50	6
Total	$n = 50$

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Procedure:

1. Draw the histogram, with class boundaries on the x-axis and frequencies on the y-axis.
2. Identify the modal class, which is the tallest bar.
3. Join the top right corner of the modal bar to the top right corner of the bar before it (the preceding bar).
4. Join the top left corner of the modal bar to the top left corner of the bar after it (the succeeding bar).
5. Mark the point where these two lines cross.
6. Draw a vertical line from the crossing point down to the x-axis. The value where it meets the x-axis is the mode.

Here the modal class is 30–40 (frequency 16). The two lines are drawn as follows.

- Line 1 joins $(40, 16)$, the top right corner of the modal bar, to $(30, 15)$, the top right corner of the preceding bar 20–30.
- Line 2 joins $(30, 16)$, the top left corner of the modal bar, to $(40, 6)$, the top left corner of the succeeding bar 40–50.

The lines cross at about (30.91, 15.09), so

$$\text{Mode} \approx 30.91$$

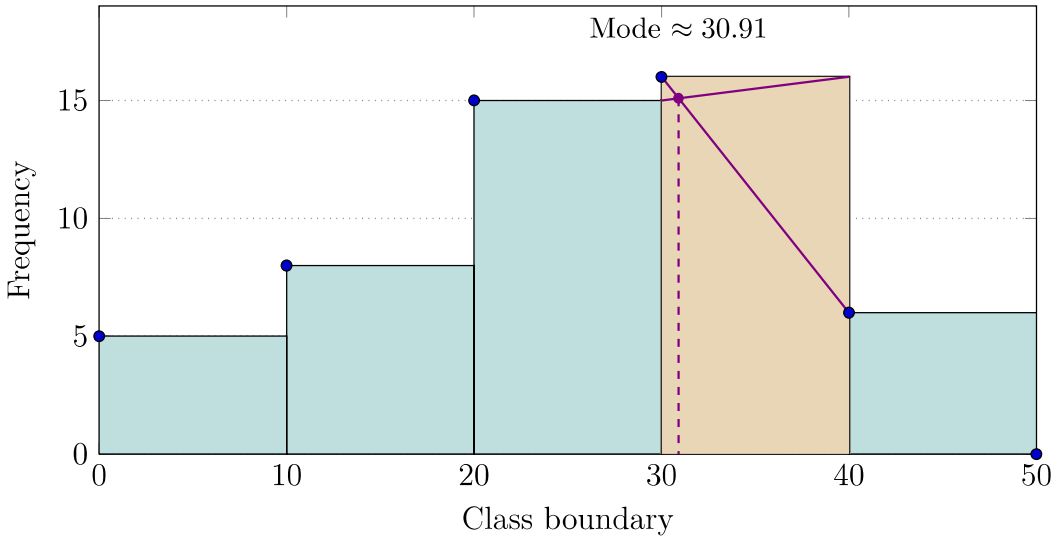


Figure 13: Mode from the histogram (the shaded bar is the modal class)

Why the method works

Let $\Delta_1 = f_1 - f_0 = 1$ and $\Delta_2 = f_1 - f_2 = 10$. Measure the distance d from the left boundary L of the modal class to the crossing point. By similar triangles, the two lines meet where

$$\frac{d}{h-d} = \frac{\Delta_1}{\Delta_2} \quad \Longrightarrow \quad d = \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

Hence the graphical mode is $L + d$, which is the same as the formula

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h = 30 + \frac{1}{11} \times 10 \approx 30.91$$

Note:

- The graphical mode is an estimate. Its accuracy depends on how carefully the graph is drawn, while the formula gives the exact interpolated value.
- The method assumes that the classes are continuous (exclusive) and have equal widths. For unequal widths, plot the frequency density $\left(\frac{f}{h}\right)$ on the y-axis.
- The mode lies closer to the neighbor with the higher frequency. Here the preceding bar (15) is much taller than the succeeding bar (6), so the mode is only slightly above 30.

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4.3.5 Merits of mode

1. It is easy to understand and calculate.
2. It is not affected by extreme values (outliers).
3. It can be calculated for open-ended classes.
4. It can be located graphically, from the histogram.
5. It can be used for qualitative data, where the mean and median cannot be used.

6. It is an actual value of the data set (for ungrouped and discrete data).

4.3.6 Demerits of mode

1. It is not rigidly defined, since a data set may have no mode or more than one mode.
2. It is not based on all observations.
3. It is not capable of further algebraic treatment.
4. It is greatly affected by sampling fluctuations.
5. For grouped data, its value depends on the choice of class intervals.
6. It is not suitable when the data have no repeated values.

4.4. Quartiles, Deciles and Percentiles

Positional measures are values that divide an ordered data set into equal parts. The median divides the data into two equal parts. The measures below divide it into more parts.

- **Quartiles** (Q_1, Q_2, Q_3) divide the data into four equal parts.
- **Deciles** $(D_1, D_2, \ldots, D_9)$ divide the data into ten equal parts.
- **Percentiles** $(P_1, P_2, \ldots, P_{99})$ divide the data into one hundred equal parts.

4.4.1 Quartiles

The first quartile Q_1 has 25% of the observations below it, the second quartile Q_2 has 50% below it, and the third quartile Q_3 has 75% below it.

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Quartiles for ungrouped data

Procedure:

1. Arrange the n observations in ascending order.
2. Find the position of the quartile using

$$Q_k = \frac{k(n+1)}{4}\text{-th value}, \qquad k = 1, 2, 3$$

3. If the position is a whole number, the quartile is the value at that position. If the position has a fractional part, interpolate between the two neighboring values:

$$\text{Value} = x_{(j)} + (\text{fractional part}) \times (x_{j+1} - x_j)$$

where j is the whole-number part of the position.

Example: Find Q_1, Q_2 and Q_3 of the following data.

$$12,\ 15,\ 18,\ 20,\ 22,\ 25,\ 28,\ 30,\ 35,\ 40$$

The data are already in ascending order and $n = 10$.

$$Q_1 = \frac{1(10+1)}{4} = 2.75\text{-th value} = 15 + 0.75(18 - 15) = 17.25$$
$$Q_2 = \frac{2(10+1)}{4} = 5.5\text{-th value} = \frac{22+25}{2} = 23.5$$
$$Q_3 = \frac{3(10+1)}{4} = 8.25\text{-th value} = 30 + 0.25(35 - 30) = 31.25$$

Quartiles for grouped data

For grouped data of a continuous variable, the quartiles are found by interpolation, in the same way as the median.

$$Q_k = L + \frac{\frac{kn}{4} - f_c}{f_m} \times h, \qquad k = 1, 2, 3$$

where

- L = lower boundary of the quartile class,
- $n = \sum f_i$ = total frequency,
- f_c = cumulative frequency of the class just before the quartile class,
- f_m = frequency of the quartile class,
- h = width of the quartile class.

The quartile class is the first class whose cumulative frequency is greater than or equal to $\frac{kn}{4}$.

One can compute the quartiles by using the following table:

Class	Frequency (f_i)	Cumulative frequency (cf)
L_1–U_1	f_1	f_1
L_2–U_2	f_2	$f_1 + f_2$
$\vdots$	$\vdots$	$\vdots$
L_k–U_k	f_k	n
Total	$\sum_{i=1}^{k} f_i = n$	

Example: Find Q_1 and Q_3 of the following distribution.

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Class	Frequency (f)	Cumulative frequency (cf)
0–10	5	5
10–20	8	13
20–30	15	28
30–40	16	44
40–50	6	50
Total	$n = 50$	

For Q_1: $\frac{n}{4} = 12.5$. The first cumulative frequency that is at least 12.5 is 13, so the Q_1 class is 10–20. Here $L = 10$, $f_c = 5$, $f_m = 8$, $h = 10$.

$$Q_1 = 10 + \frac{12.5 - 5}{8} \times 10 = 10 + 9.375 = 19.375$$

For Q_3: $\frac{3n}{4} = 37.5$. The first cumulative frequency that is at least 37.5 is 44, so the Q_3 class is 30–40. Here $L = 30$, $f_c = 28$, $f_m = 16$, $h = 10$.

$$Q_3 = 30 + \frac{37.5 - 28}{16} \times 10 = 30 + 5.9375 = 35.9375$$

Also, Q_2 is the median, which was found earlier to be 28.

4.4.2 Deciles

The deciles divide the data into ten equal parts. The k-th decile D_k has $\frac{k}{10}$ of the observations below it. For example, D_1 has 10% below it and D_5 has 50% below it.

Deciles for ungrouped data

$$D_k = \frac{k(n+1)}{10}\text{-th value}, \qquad k = 1, 2, \ldots, 9$$

Arrange the data in ascending order. If the position has a fractional part, interpolate between the two neighboring values, as for the quartiles.

Example: For the data $12, 15, 18, 20, 22, 25, 28, 30, 35, 40$ (here $n = 10$), find D_3 and D_7.

$$D_3 = \frac{3(10+1)}{10} = 3.3\text{-th value} = 18 + 0.3(20 - 18) = 18.6$$

$$D_7 = \frac{7(10+1)}{10} = 7.7\text{-th value} = 28 + 0.7(30 - 28) = 29.4$$

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Deciles for grouped data

$$D_k = L + \frac{\frac{kn}{10} - f_c}{f_m} \times h, \qquad k = 1, 2, \ldots, 9$$

The symbols have the same meanings as for the quartiles. The decile class is the first class whose cumulative frequency is greater than or equal to $\frac{kn}{10}$.

Example: For the distribution in the quartile example, find D_4.
Here $\frac{4n}{10} = 20$. The first cumulative frequency that is at least 20 is 28, so the D_4 class is 20–30. Here $L = 20$, $f_c = 13$, $f_m = 15$, $h = 10$.

$$D_4 = 20 + \frac{20 - 13}{15} \times 10 = 20 + 4.67 = 24.67$$

4.4.3 Percentiles

The percentiles divide the data into one hundred equal parts. The k-th percentile P_k has $k\%$ of the observations below it. For example, a student at the 90th percentile scored higher than 90% of the students.

Percentiles for ungrouped data

$$P_k = \frac{k(n+1)}{100}\text{-th value}, \qquad k = 1, 2, \ldots, 99$$

Arrange the data in ascending order. If the position has a fractional part, interpolate between the two neighboring values.

Example: For the data $12, 15, 18, 20, 22, 25, 28, 30, 35, 40$ (here $n = 10$), find P_{90}.

$$P_{90} = \frac{90(10+1)}{100} = 9.9\text{-th value} = 35 + 0.9(40 - 35) = 39.5$$

Percentiles for grouped data

$$P_k = L + \frac{\frac{kn}{100} - f_c}{f_m} \times h, \qquad k = 1, 2, \ldots, 99$$

The symbols have the same meanings as before. The percentile class is the first class whose cumulative frequency is greater than or equal to $\frac{kn}{100}$.

Example: For the distribution in the quartile example, find P_{90}.
Here $\frac{90n}{100} = 45$. The first cumulative frequency that is at least 45 is 50, so the P_{90} class is 40–50. Here $L = 40$, $f_c = 44$, $f_m = 6$, $h = 10$.

$$P_{90} = 40 + \frac{45 - 44}{6} \times 10 = 40 + 1.67 = 41.67$$

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Relationship among quartiles, deciles and percentiles

These measures are connected, because they are all points that split the same ordered data.

$$Q_1 = P_{25}, \qquad Q_2 = D_5 = P_{50} = \text{Median}, \qquad Q_3 = P_{75}$$
$$D_k = P_{10k}, \qquad k = 1, 2, \ldots, 9$$

Note:

- The data must always be arranged in ascending order before an ungrouped quartile, decile or percentile is found.
- Different textbooks and software packages use slightly different position rules for ungrouped data. This section uses $\frac{k(n+1)}{4}$, $\frac{k(n+1)}{10}$ and $\frac{k(n+1)}{100}$, so another source may give a slightly different value for the same data.
- For grouped data the classes must be continuous (exclusive). If they are inclusive, first convert them to class boundaries.
- The grouped results are estimates, because the values inside each class are assumed to be spread evenly.
- The difference $Q_3 - Q_1$ is called the interquartile range and is a measure of spread.

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Locating Quartiles, Deciles and Percentiles Graphically

The quartiles, deciles and percentiles of grouped data of a continuous variable can be located from the "less than" ogive, in the same way as the median. The same distribution as before is used here.

Less than	Cumulative frequency
0	0
10	5
20	13
30	28
40	44
50	50

Procedure

1. Draw the "less than" ogive, with the upper class boundaries on the x-axis and the cumulative frequencies on the y-axis.

2. Find the required height on the y-axis:

$$Q_k : \frac{kn}{4}, \qquad D_k : \frac{kn}{10}, \qquad P_k : \frac{kn}{100}$$

3. Draw a horizontal line from this height until it meets the ogive.

4. From that point draw a vertical line down to the x-axis. The value where it meets the x-axis is the required quartile, decile or percentile.

4.4.4 Locating the quartiles

Here $n = 50$, so the heights are

$$\frac{n}{4} = 12.5, \qquad \frac{2n}{4} = 25, \qquad \frac{3n}{4} = 37.5$$

Reading the ogive in Figure 14 gives

$$Q_1 \approx 19.4, \qquad Q_2 = 28, \qquad Q_3 \approx 35.9$$

which agree with the values found from the formulas.

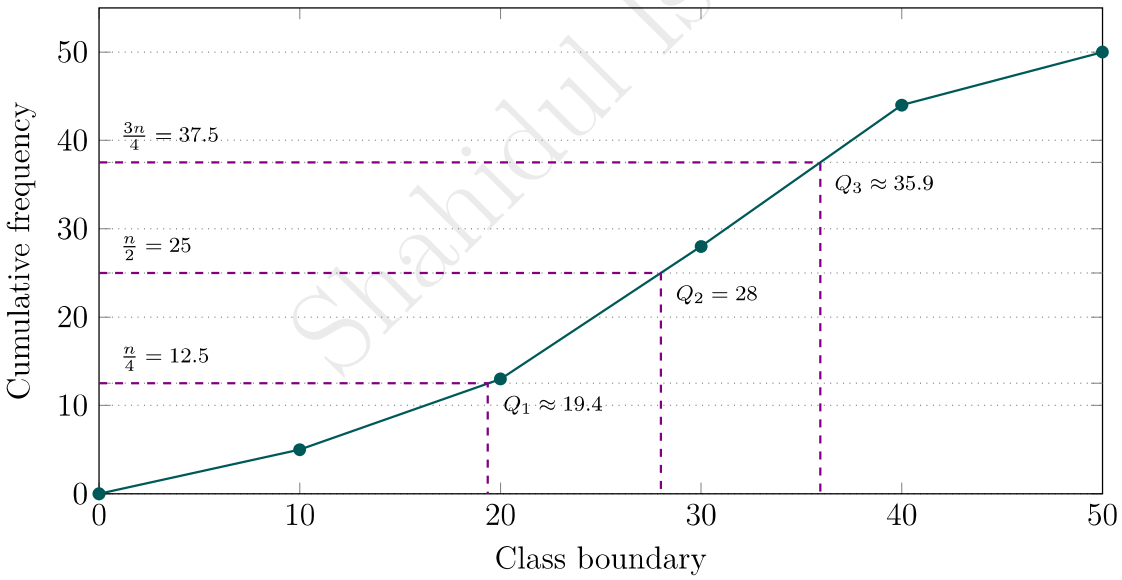


Figure 14: Quartiles from the "less than" ogive

4.4.5 Locating a decile and a percentile

The method is the same, and only the height on the y-axis changes.

- For D_4, the height is $\dfrac{4n}{10} = 20$. The ogive gives $D_4 \approx 24.7$.
- For P_{90}, the height is $\dfrac{90n}{100} = 45$. The ogive gives $P_{90} \approx 41.7$.

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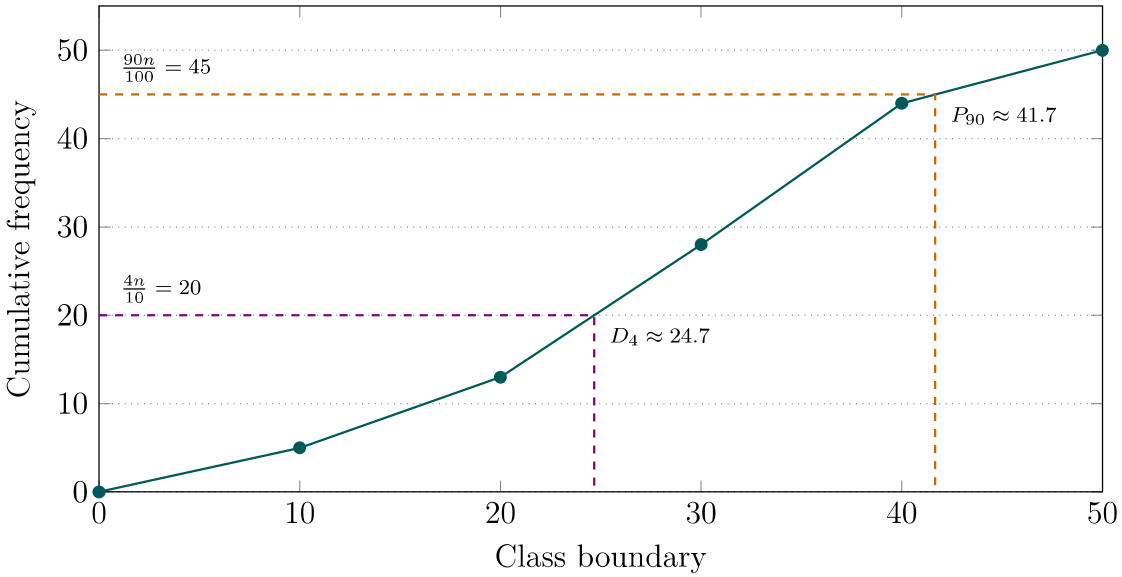


Figure 15: Decile D_4 and percentile P_{90} from the "less than" ogive

Note:

- The graphical values are estimates, and their accuracy depends on how carefully the graph is drawn. The formula gives the exact interpolated value.
- One ogive can be used to read off every quartile, decile and percentile, so it only has to be drawn once.
- The quartiles, deciles and percentiles are related by $Q_2 = D_5 = P_{50} =$ Median, so the median found in the earlier section is the same point as Q_2 here.
- Alternatively, a second y-axis can be marked in percentages from 0% to 100%. Then P_k is read directly at the height of k%, without calculating $\dfrac{kn}{100}$.
- The classes must be continuous (exclusive). If they are inclusive, first convert them to class boundaries.

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4.5. Geometric mean

The geometric mean (G.M.) of n positive observations is the n-th root of their product. It is used for averaging ratios, rates of change and growth factors.

4.5.1 Geometric mean for ungrouped data

For positive observations $x_1, x_2, \ldots, x_n$,

$$G = (x_1 \cdot x_2 \cdots x_n)^{1/n}$$

Taking logarithms, this can be computed as

$$\log G = \frac{1}{n}\sum_{i=1}^{n} \log x_i \quad \Longrightarrow \quad G = \text{antilog}\left(\frac{\sum_{i=1}^{n} \log x_i}{n}\right)$$

Example: Find the geometric mean of $2, 4, 8$.

$$G = (2 \times 4 \times 8)^{1/3} = 64^{1/3} = 4$$

Geometric mean from grouped data of continuous variable

For values (or class midpoints) x_i with frequencies f_i,

$$G = \text{antilog}\left(\frac{\sum_{i=1}^{k} f_i \log x_i}{\sum_{i=1}^{k} f_i}\right)$$

One can compute G by using the following table:

Class	Frequency (f_i)	Midpoint (x_i)	$\log x_i$	$f_i \log x_i$
L_1–U_1	f_1	x_1	$\log x_1$	$f_1 \log x_1$
L_2–U_2	f_2	x_2	$\log x_2$	$f_2 \log x_2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
L_k–U_k	f_k	x_k	$\log x_k$	$f_k \log x_k$
Total	$\sum f_i = n$			$\sum f_i \log x_i$

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Application of the geometric mean

- It is used to find the average rate of growth, such as population growth, inflation or investment returns.
- It is used for averaging ratios, percentages and index numbers.
- It is the correct average when quantities multiply over time, rather than add.

Example: An investment grows by 10%, 20% and 50% in three successive years. The growth factors are 1.10, 1.20 and 1.50, so

$$G = (1.10 \times 1.20 \times 1.50)^{1/3} = (1.98)^{1/3} \approx 1.2557$$

The average growth rate is about 25.57% per year. The arithmetic mean of the rates, 26.67%, overstates the true average growth.

Merits and demerits of the Geometric Mean

- **Merits:** It is rigidly defined, based on all observations, capable of further algebraic treatment, and gives less weight to extreme large values than the arithmetic mean.
- **Demerits:** It is hard to understand and calculate, and it cannot be found if any value is zero or negative.

4.5.2 Harmonic mean

The harmonic mean (H.M.) of n non-zero observations is the reciprocal of the arithmetic mean of their reciprocals. It is used for averaging rates and speeds.

Harmonic mean for ungrouped data

For non-zero observations $x_1, x_2, \ldots, x_n$,

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Example: Find the harmonic mean of $2, 3, 6$.

$$H = \frac{3}{\frac{1}{2} + \frac{1}{3} + \frac{1}{6}} = \frac{3}{1} = 3$$

Harmonic mean from grouped data of continuous variable

For values (or class midpoints) x_i with frequencies f_i,

$$H = \frac{\sum_{i=1}^{k} f_i}{\sum_{i=1}^{k} \frac{f_i}{x_i}} = \frac{n}{\sum_{i=1}^{k} \frac{f_i}{x_i}}$$

One can compute H by using the following table:

Class	Frequency (f_i)	Midpoint (x_i)	$\frac{1}{x_i}$	$\frac{f_i}{x_i}$
L_1–U_1	f_1	x_1	$\frac{1}{x_1}$	$\frac{f_1}{x_1}$
L_2–U_2	f_2	x_2	$\frac{1}{x_2}$	$\frac{f_2}{x_2}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
L_k–U_k	f_k	x_k	$\frac{1}{x_k}$	$\frac{f_k}{x_k}$
Total	$\sum f_i = n$			$\sum \frac{f_i}{x_i}$

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Application of the harmonic mean

- It is used to find the average speed when equal distances are covered at different speeds.
- It is used for averaging rates, such as price per unit, when equal amounts of money are spent.
- It is the correct average when the quantities are ratios of the form "per unit of something".

Example: A car travels from A to B at 60 km/h and returns from B to A at 40 km/h. The distance is the same each way, so the average speed is the harmonic mean:

$$H = \frac{2}{\frac{1}{60} + \frac{1}{40}} = \frac{2}{\frac{1}{24}} = 48 \text{ km/h}$$

The arithmetic mean, 50 km/h, is wrong here, because the car spends more time at the lower speed.

Merits and demerits of the harmonic mean

- **Merits:** It is rigidly defined, based on all observations, capable of further algebraic treatment, and gives the most weight to small values.
- **Demerits:** It is hard to understand and calculate, it cannot be found if any value is zero, and it is strongly affected by very small values.

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4.5.3 Relationship Among A.M., G.M. and H.M.

(i) For positive observations that are not all equal,

$$\text{A.M.} > \text{G.M.} > \text{H.M.}$$

and the three are equal only when all observations are equal.

(ii) For two positive numbers a and b, also

$$G^2 = \text{A.M.} \times \text{H.M.}$$

Example: For $a = 2$ and $b = 8$,

$$\text{A.M.} = 5, \qquad G = \sqrt{2 \times 8} = 4, \qquad H = \frac{2}{\frac{1}{2} + \frac{1}{8}} = 3.2$$

so $5 > 4 > 3.2$, and $G^2 = 16 = 5 \times 3.2$.

4.6. Some Simple Theorems and Problems

Theorem 1. For two positive non-zero quantities,

$$\text{A.M.} \geq \text{G.M.} \geq \text{H.M.}$$

Here A.M. = Arithmetic mean, G.M. = Geometric mean, and H.M. = Harmonic mean.

Solution. Suppose x_1 and x_2 are two positive and non-zero quantities. Then

$$\text{A.M.} = \frac{x_1 + x_2}{2}, \quad \text{G.M.} = \sqrt{x_1 \times x_2}, \quad \text{and} \quad \text{H.M.} = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}}.$$

Here $\left(\sqrt{x_1} - \sqrt{x_2}\right)^2 \geq 0$ since x_1 and x_2 are positive.

$$\begin{aligned}
&\text{or, } x_1 + x_2 - 2\sqrt{x_1 \times x_2} \geq 0 \\
&\text{or, } x_1 + x_2 \geq 2\sqrt{x_1 \times x_2} \\
&\text{or, } \frac{x_1 + x_2}{2} \geq \sqrt{x_1 \times x_2}
\end{aligned}$$

Hence,

$$\text{A.M.} \geq \text{G.M.} \tag{1}$$

Again,

$$\begin{aligned}
&\left(\frac{1}{\sqrt{x_1}} - \frac{1}{\sqrt{x_2}}\right)^2 \geq 0 \\
&\text{or, } \frac{1}{x_1} + \frac{1}{x_2} - \frac{2}{\sqrt{x_1 \times x_2}} \geq 0 \\
&\text{or, } \frac{1}{x_1} + \frac{1}{x_2} \geq \frac{2}{\sqrt{x_1 \times x_2}} \\
&\text{or, } \sqrt{x_1 \times x_2} \geq \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}}
\end{aligned}$$

Hence,

$$\text{G.M.} \geq \text{H.M.} \tag{2}$$

From (1) and (2), we have:

$$\text{A.M.} \geq \text{G.M.} \geq \text{H.M.}$$

Theorem 2. For two non-zero and positive quantities,

$$\text{G.M.} = \sqrt{\text{A.M.} \times \text{H.M.}}$$

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Proof. Suppose x_1 and x_2 are two positive and non-zero quantities. Then

$$\text{A.M.} = \frac{x_1 + x_2}{2}, \quad \text{G.M.} = \sqrt{x_1 \times x_2}, \quad \text{and} \quad \text{H.M.} = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}}.$$

Now,

$$\begin{aligned}
\text{A.M.} \times \text{H.M.} &= \frac{x_1 + x_2}{2} \times \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}} \\
&= \frac{x_1 + x_2}{2} \times \frac{2}{\frac{x_1 + x_2}{x_1 \times x_2}} \\
&= \frac{x_1 + x_2}{2} \times \frac{2(x_1 \times x_2)}{x_1 + x_2} \\
&= x_1 \times x_2 = (\sqrt{x_1 \times x_2})^2 = (\text{G.M.})^2
\end{aligned}$$

Hence,

$$\text{G.M.} = \sqrt{\text{A.M.} \times \text{H.M.}}$$

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Problem 1. Suppose the arithmetic mean and harmonic mean of two positive quantities are 8 and 2. Find their geometric mean.

Solution. Here A.M. = 8 and H.M. = 2.

We know,

$$\begin{aligned}
\text{A.M.} \times \text{H.M.} &= (\text{G.M.})^2 \\
8 \times 2 &= (\text{G.M.})^2 \\
16 &= (\text{G.M.})^2
\end{aligned}$$

Hence,

$$\text{G.M.} = \sqrt{16} = 4.$$

Problem 2. Suppose the geometric mean and harmonic mean of two positive quantities are $4\sqrt{3}$ and 6. Find their arithmetic mean.

Solution. Here, G.M. $= 4\sqrt{3}$ and H.M. = 6.

We know,

$$\begin{aligned}
\text{A.M.} \times \text{H.M.} &= (\text{G.M.})^2 \\
\text{A.M.} \times 6 &= (4\sqrt{3})^2 = 48
\end{aligned}$$

Hence,

$$\text{A.M.} = \frac{48}{6} = 8.$$

Problem 3. For n natural numbers, arithmetic mean is equal to $(n+1)/2$.

Solution. The first n natural numbers are $1, 2, 3, \ldots, n$.

Then

$$\text{A.M.} = \frac{1+2+3+\cdots+n}{n} = \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}.$$

Problem 4. Find the arithmetic mean of $1, 2, 3, \ldots, 20$.

Solution. Here $n = 20$.

$$\text{Hence A.M.} = \frac{20+1}{2} = 10.5.$$

Problem 5. For n natural numbers, median is equal to $(n+1)/2$.

Solution. Let $1, 2, 3, \ldots, n$ be n natural numbers. n may be odd or even number.

First let us take n is odd number. Suppose $n = 2m+1$, then median will be $(m+1)$th value. In this case $m = (n-1)/2$.

By definition,

$$\text{Median} = \left[\frac{n-1}{2} + 1\right] \text{th value} = \frac{n+1}{2}.$$

Now, let us take n is even. Then $n = 2m$. In this case $m = n/2$.

By definition, median will be mean of mth value and $(m+1)$th value.

Then,

$$\text{Median} = \frac{\frac{n}{2} + \frac{n}{2} + 1}{2} = \frac{n+1}{2}.$$

It is seen that in both the cases median is $\frac{n+1}{2}$.

Note. For first n natural numbers mean and median are same.

Shahidul Islam